

Disk, disk, sleuth!

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Use **srch_strings** from the sleuthkit and some terminal-fu to find a flag in this disk image.

Hints

- Have you ever used `file` to determine what a file was?
 - Relevant terminal-fu in picoGym: <https://learn.cylabacademy.org/library/85>
 - Mastering this terminal-fu would enable you to find the flag in a single command: <https://learn.cylabacademy.org/library/48>
 - Using your own computer, you could use `qemu` to boot from this disk!
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Tools Used

- `7z` – Extract compressed files
 - `file` – Determine file type
 - `srch_strings` – Extract printable character sequences from binary files (Sleuth Kit)
 - `grep` – Filter output for specific patterns
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Complete Solution

Step 1: Download and Extract

Download the disk image named `dds1-alpine.flag.img.gz` from the challenge page.

Extract the compressed disk image:

```
7z x dds1-alpine.flag.img.gz
```

This decompresses the gzipped file and creates the raw disk image `dds1-alpine.flag.img`.

Step 2: Basic File Inspection

Verify the file type and understand its structure:

```
file dds1-alpine.flag.img
```

Expected output:

```
dds1-alpine.flag.img: DOS/MBR boot sector; partition 1 : ID=0x83, active, start-CHS  
(0x0,32,33), end-CHS (0x10,81,1), startsector 2048, 260096 sectors
```

The output confirms this is a **DOS/MBR boot sector** with a Linux partition (ID `0x83`).

Step 3: Search for the Flag with `srch_strings`

The challenge specifically mentions using `srch_strings` from **The Sleuth Kit**. This tool is similar to the standard `strings` command but offers additional options for forensic analysis.

Search the entire disk image for any string containing `picoCTF` :

```
srch_strings -f dds1-alpine.flag.img | grep 'picoCTF'
```

Output:

```
dds1-alpine.flag.img:  SAY picoCTF{<REDACTED>}
```

Key finding: The flag appears directly in the output!

Step 4: Alternative – Scan the Entire File

The `-a` option scans the entire file, not just the data section (equivalent to the standard `strings -a`):

```
srch_strings -a dds1-alpine.flag.img | grep 'picoCTF'
```

Output:

```
SAY picoCTF{<REDACTED>}
```

 Understanding the `srch_strings` Options

Option	Description
-a	Scan the entire file, not just the data section
-f	Print the name of the file before each string

 Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, the `srch_strings` command will reveal the complete flag.

 Key Concepts

Concept	Description
Strings Extraction	A powerful technique for finding readable text embedded in binary files.
<code>srch_strings</code>	A Sleuth Kit tool similar to <code>strings</code> but with forensic-specific options.

Concept	Description
Binary Analysis	Sometimes flags are directly visible as strings within binary data without needing specialized tools.
Disk Image	A raw sector-by-sector copy of a storage device, which can be searched directly with <code>strings</code> or <code>srch_strings</code> .

Pro Tips

1. **Start with `strings` or `srch_strings`** – Before diving deep into forensic analysis, a quick scan for the flag prefix can save hours of work.
2. **Know the difference between `strings` and `srch_strings`** – `srch_strings` is part of The Sleuth Kit and offers additional options useful for forensic investigations.
3. **Use `-a` to scan everything** – Some data may be stored outside the default data sections. The `-a` option ensures you don't miss anything.
4. **The flag prefix is your best friend** – `grep "picoCTF"` is the fastest way to locate the flag in almost any challenge.
5. **Don't overcomplicate** – Sometimes the simplest solution is the correct one. No need to mount the disk image or analyze partitions when the flag appears in a simple string search.